Additional Consensus Driven Pseudo-Labeling for Long-Tailed Multi-view Semi-Supervised Classification

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This document complements the main paper with the experimental results that are not included in the main body. Section I reports the complete Accuracy, G-mean, and Specificity tables for the same 84 experimental settings as the Macro-F1 table in the main paper, and Section II reports the full label-ratio results behind the label-ratio figure in the main paper. Sections III–VIII provide the additional figures for the radar charts, pseudo-label analysis, class-group comparison, parameter sensitivity, and representation visualization, together with the extended results on the clinical IGF-1 application. All conventions, datasets, comparison methods, and evaluation metrics follow the definitions in Section IV of the main paper. Throughout, *C*/*I*/*A*/*U* denote the consistent, inverse, arbitrary, and uniform unlabeled distributions, respectively, and IR and LR denote the imbalance ratio and the labeling ratio.

I. Additional Full Results Tables

The main paper reports the complete Macro-F1 results across all seven benchmarks under $IR \in \{0.1, 0.2, 0.3\}$ and the four unlabeled distributions at $LR = 0.09$. This section provides the corresponding full tables for the three remaining evaluation metrics: Accuracy (Table I), G-mean (Table II), and macro-averaged Specificity (Table III). In each table, the best mean is pink and shown in bold, while the second-best mean is blue and underlined. Values are reported as mean $\pm$ standard deviation in percent. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20; DSRL additionally ran out of memory on the three OutScene uniform settings. Missing runs are excluded from the ranking.

II. Full Label-Ratio Results

The main paper illustrates the impact of the labeling ratio (LR) on Macro-F1 across CIFAR-100, GRAZ02, NoisyMNIST, and OutScene. Tables IV–VII provide the complete results for four evaluation metrics at $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$ under the consistent long-tailed setting ($IR = 0.1$). Results are presented as mean $\pm$ standard deviation across three random seeds, with missing results due to memory limitations denoted by "$-$". Method names include the selected learning rates, and CDPL is denoted as *Ours*. CDPL consistently achieves the highest Macro-F1 across all datasets and labeling ratios, with increasingly pronounced gains over the strongest baseline as the labeling ratio decreases.

III. Additional Radar Charts at $IR = 0.2$ and $IR = 0.3$

The main paper visualizes the Accuracy, G-mean, and Specificity of all compared methods at $IR = 0.1$. Figures 1 and 2 extend the comparison to $IR = 0.2$ and $IR = 0.3$, respectively, using the same radar representation. Consistent with the main paper, the CDPL polygon remains the outermost on all six axes and in all panels, confirming that the advantage of CDPL is not specific to the most severe imbalance setting.

IV. Additional Pseudo-Label Analysis

The main paper reports the pseudo-label error rate and utilization rate on CIFAR-100, NoisyMNIST, and YouTube-Face20, together with the row-normalized confusion matrices of SimPro and CDPL from epoch 100 to 500. This section supplements those results with (i) the pseudo-label statistics on CIFAR-10 (Figure 3) and (ii) the row-normalized pseudo-label confusion matrices of SimPro and CDPL on NoisyMNIST at $LR = 0.09$ and $IR = 0.1$ (Figure 4).

V. Additional Class-Group Results

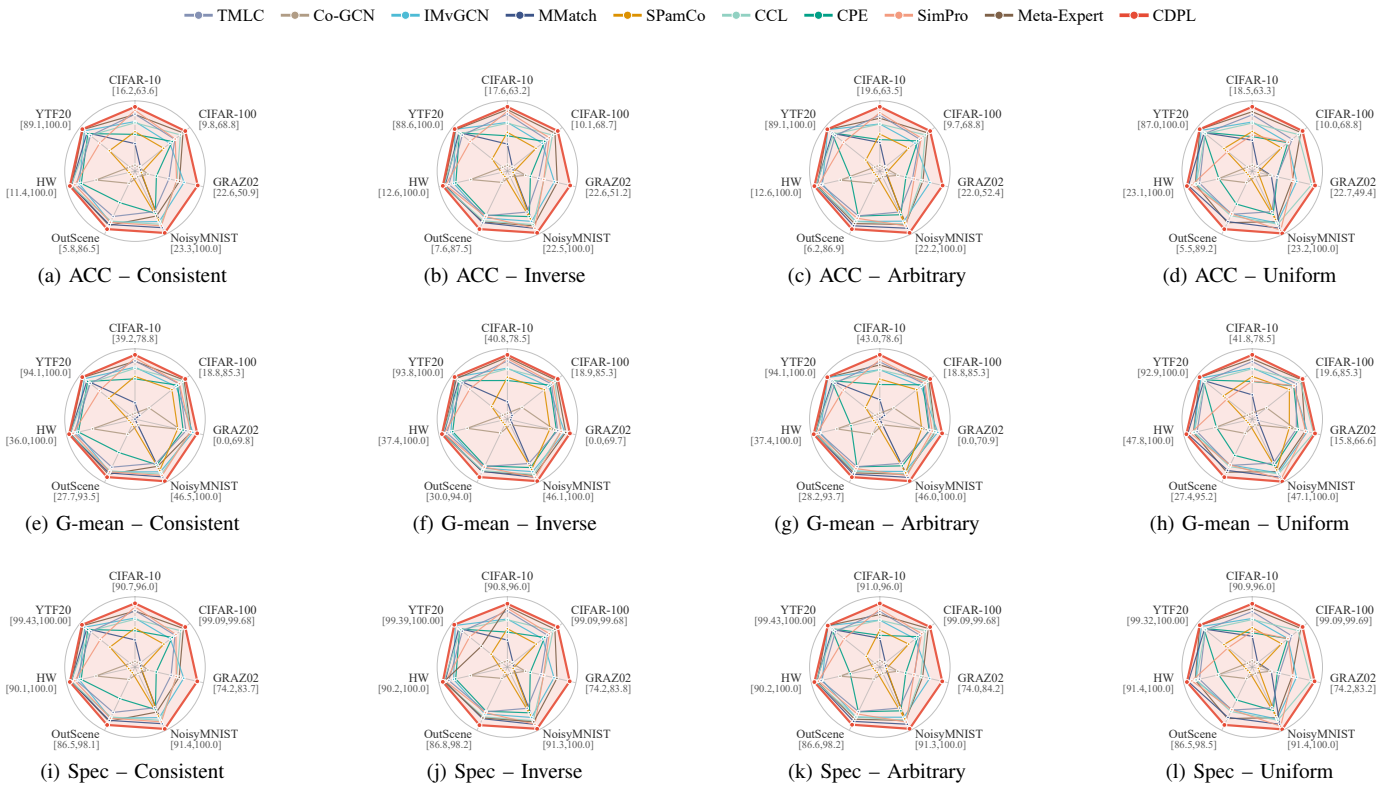
The main paper evaluates Macro-F1 across head, medium, and tail classes on CIFAR-10, NoisyMNIST, CIFAR-100, and YouTubeFace20. Figure 5 extends this analysis to Handwritten and OutScene, where CDPL consistently achieves the highest Macro-F1 across all class groups, with the most pronounced gains on tail classes.

VI. Additional Parameter Sensitivity

The main paper analyzes the sensitivity of CDPL to the trade-off coefficients λ_1, λ_2, and λ_3 on NoisyMNIST and OutScene. Figure 6 extends this analysis to Handwritten and GRAZ02. The results are consistent with the main paper: performance is relatively insensitive to λ_1 and is best when λ_2 is near 10 and λ_3 is near 1.

VII. Comprehensive t-SNE Visualization of CDPL Representations

The main paper visualizes nine methods on the YouTube-Face20 test set. Figures 7–8 further examine CDPL's fused representations on the NoisyMNIST and YouTubeFace20 datasets under four unlabeled distributions (Consistent, Inverse, Arbitrary, and Uniform), covering both training and test sets ($LR = 0.09$, $IR = 0.1$). Across all settings, CDPL produces

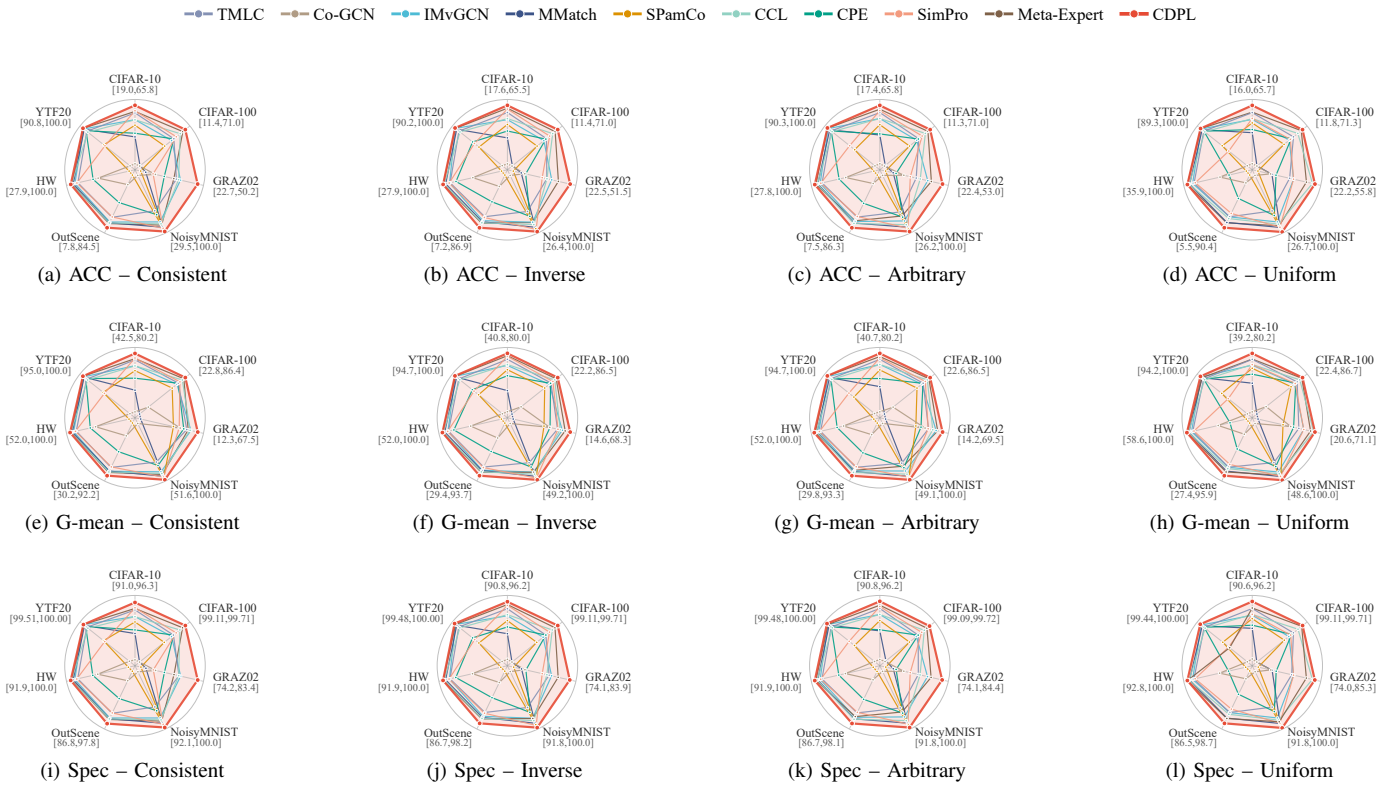


(a) ACC – Consistent (b) ACC – Inverse (c) ACC – Arbitrary (d) ACC – Uniform

(e) G-mean – Consistent (f) G-mean – Inverse (g) G-mean – Arbitrary (h) G-mean – Uniform

(i) Spec – Consistent (j) Spec – Inverse (k) Spec – Arbitrary (l) Spec – Uniform

Fig. 1. Accuracy, G-mean, and Specificity of all compared methods on six datasets under the four unlabeled distributions at $IR = 0.2$ and $LR = 0.09$.



(a) ACC – Consistent (b) ACC – Inverse (c) ACC – Arbitrary (d) ACC – Uniform

(e) G-mean – Consistent (f) G-mean – Inverse (g) G-mean – Arbitrary (h) G-mean – Uniform

(i) Spec – Consistent (j) Spec – Inverse (k) Spec – Arbitrary (l) Spec – Uniform

Fig. 2. Accuracy, G-mean, and Specificity of all compared methods on six datasets under the four unlabeled distributions at $IR = 0.3$ and $LR = 0.09$.

TABLE I

ACCURACY (%) ON SEVEN DATASETS AT $LR = 0.09$ AND $IR \in \{0.1, 0.2, 0.3\}$ UNDER FOUR UNLABELED DISTRIBUTIONS. VALUES ARE MEAN ± REPORTED VARIATION. THE BEST AND SECOND-BEST MEANS ARE MARKED BY PINK/BOLD AND BLUE/UNDERLINED ENTRIES, RESPECTIVELY. THE Δ ROW REPORTS CDPL MINUS THE STRONGEST COMPETING METHOD (PERCENTAGE POINTS); POSITIVE/NEGATIVE DIFFERENCES ARE BOLD RED/BLUE.

Dataset	Method	$IR = 0.1$				$IR = 0.2$				$IR = 0.3$			
		C	I	A	U	C	I	A	U	C	I	A	U
CIFAR-100	CCL	52.11±0.006	51.65±0.005	51.23±0.002	52.55±0.006	58.61±0.004	58.29±0.002	58.58±0.007	59.25±0.010	60.49±0.004	60.54±0.004	60.79±0.008	61.16±0.015
	CPE	42.91±0.009	41.02±0.009	43.70±0.003	44.01±0.009	49.75±0.003	49.91±0.011	50.50±0.010	48.20±0.010	53.37±0.008	53.10±0.002	53.18±0.021	54.02±0.004
	SimPro	44.87±0.014	47.47±0.005	47.46±0.018	40.84±0.036	53.04±0.016	54.76±0.009	54.22±0.012	48.78±0.011	56.27±0.014	57.19±0.010	57.28±0.011	53.07±0.017
	Meta-Expert	52.29±0.030	49.78±0.018	52.29±0.016	52.24±0.003	61.37±0.005	60.84±0.005	61.29±0.009	61.96±0.006	63.30±0.001	63.21±0.013	63.61±0.006	64.91±0.004
	Co-GCN	12.47±0.084	13.03±0.082	12.66±0.083	12.65±0.081	14.72±0.105	15.06±0.099	14.63±0.102	14.93±0.095	16.41±0.109	16.40±0.108	16.31±0.109	16.84±0.113
	IMvGCN	40.43±0.003	41.39±0.004	41.21±0.004	42.79±0.002	47.22±0.007	47.30±0.001	47.52±0.003	48.20±0.001	51.17±0.004	50.97±0.002	51.44±0.003	51.91±0.004
	MMatch	13.72±0.001	14.07±0.001	14.11±0.000	14.00±0.000	15.64±0.000	15.46±0.000	15.62±0.004	15.86±0.003	16.86±0.005	16.81±0.001	17.23±0.003	16.91±0.009
	SPamCo	23.89±0.127	32.35±0.003	31.97±0.003	32.74±0.010	40.59±0.004	40.59±0.004	40.59±0.004	41.23±0.003	43.61±0.004	43.61±0.004	43.61±0.004	46.28±0.002
	TMLC	47.11±0.000	47.34±0.001	47.41±0.000	46.89±0.001	52.62±0.000	52.35±0.000	52.71±0.001	52.68±0.005	53.57±0.001	53.77±0.000	53.56±0.000	56.48±0.001
	CDPL	**58.94±0.003**	**58.94±0.003**	**58.94±0.003**	**59.61±0.002**	**63.82±0.004**	**63.82±0.004**	**63.82±0.004**	**63.89±0.004**	**66.00±0.006**	**66.00±0.006**	**66.00±0.006**	**66.26±0.002**
	Δ	**+6.65**	**+7.29**	**+6.65**	**+7.06**	**+2.45**	**+2.98**	**+2.53**	**+1.93**	**+2.70**	**+2.79**	**+2.39**	**+1.35**
CIFAR-10	CCL	41.83±0.022	43.01±0.033	42.83±0.030	43.17±0.021	48.81±0.030	48.73±0.030	49.05±0.033	49.16±0.029	52.26±0.044	52.48±0.046	52.69±0.048	52.17±0.047
	CPE	36.69±0.014	35.26±0.018	36.10±0.009	36.96±0.005	41.00±0.011	40.52±0.004	39.30±0.016	40.51±0.012	43.32±0.009	44.01±0.011	42.08±0.006	44.48±0.016
	SimPro	53.71±0.042	54.07±0.028	53.69±0.031	53.74±0.032	57.28±0.039	56.46±0.033	56.15±0.029	41.50±0.143	58.29±0.026	59.00±0.032	58.12±0.024	52.34±0.095
	Meta-Expert	43.55±0.139	54.27±0.040	45.28±0.138	47.01±0.149	54.41±0.061	57.53±0.028	52.66±0.092	56.40±0.041	57.85±0.057	59.54±0.031	59.09±0.027	56.76±0.066
	Co-GCN	19.70±0.057	21.99±0.034	20.53±0.046	20.82±0.051	20.23±0.061	21.47±0.051	23.28±0.036	22.24±0.039	22.96±0.043	21.61±0.051	21.50±0.053	20.22±0.054
	IMvGCN	45.44±0.002	44.79±0.004	44.76±0.004	46.31±0.002	49.64±0.004	49.25±0.003	49.06±0.002	49.84±0.007	52.36±0.002	51.76±0.003	52.70±0.001	51.61±0.002
	MMatch	29.06±0.011	29.90±0.011	29.45±0.086	36.63±0.015	34.78±0.027	35.16±0.033	37.49±0.089	38.23±0.048	40.89±0.009	39.33±0.012	41.30±0.046	42.21±0.005
	SPamCo	36.21±0.004	35.63±0.002	35.62±0.002	36.70±0.000	42.78±0.005	42.78±0.005	42.78±0.005	43.81±0.001	48.67±0.005	48.46±0.005	48.46±0.005	49.11±0.004
	TMLC	53.18±0.000	52.98±0.000	52.96±0.001	52.08±0.000	54.64±0.001	54.68±0.000	54.61±0.002	54.41±0.001	56.88±0.000	56.61±0.000	56.80±0.002	56.40±0.000
	CDPL	**54.58±0.002**	**55.88±0.000**	**54.97±0.008**	**55.44±0.001**	**59.57±0.006**	**59.37±0.002**	**59.81±0.002**	**59.54±0.005**	**61.90±0.004**	**61.51±0.003**	**61.73±0.001**	**61.51±0.004**
	Δ	**+0.87**	**+1.61**	**+1.28**	**+1.70**	**+2.29**	**+1.84**	**+3.66**	**+3.14**	**+3.61**	**+1.97**	**+2.64**	**+4.75**
GRAZ02	CCL	35.75±0.018	34.27±0.015	32.80±0.006	44.62±0.047	39.92±0.008	37.90±0.011	40.32±0.015	45.83±0.038	40.19±0.005	39.25±0.008	40.46±0.006	51.75±0.008
	CPE	31.45±0.091	28.23±0.046	31.51±0.029	34.75±0.027	31.45±0.091	32.20±0.047	33.57±0.033	32.46±0.106	34.46±0.123	30.24±0.074	29.66±0.088	33.86±0.114
	SimPro	38.84±0.030	39.38±0.023	38.44±0.012	43.01±0.019	39.78±0.026	37.77±0.054	37.23±0.030	38.58±0.026	29.97±0.009	37.23±0.012	36.02±0.023	42.47±0.005
	Meta-Expert	37.23±0.034	36.42±0.013	36.42±0.031	42.88±0.002	40.99±0.033	43.41±0.023	41.13±0.058	38.31±0.057	38.31±0.023	44.09±0.008	45.43±0.043	48.79±0.023
	Co-GCN	29.03±0.033	28.16±0.048	25.47±0.037	27.55±0.033	27.96±0.038	30.24±0.040	30.04±0.035	29.64±0.038	30.91±0.037	29.84±0.014	33.13±0.025	31.18±0.045
	DSRL	29.03±0.000	27.02±0.000	25.00±0.000	35.48±0.000	31.05±0.000	31.85±0.000	25.00±0.000	33.87±0.000	29.84±0.000	25.00±0.000	29.44±0.000	37.50±0.000
	GEGCN	26.48±0.026	26.34±0.022	27.96±0.030	34.95±0.023	31.05±0.011	29.70±0.043	31.59±0.076	37.77±0.015	40.86±0.019	39.11±0.054	38.31±0.032	43.01±0.093
	IMvGCN	38.44±0.010	40.73±0.007	40.19±0.019	41.53±0.019	42.88±0.014	42.20±0.010	44.22±0.014	40.05±0.016	40.99±0.008	41.26±0.008	43.82±0.033	42.47±0.014
	MMatch	24.87±0.002	25.00±0.000	25.00±0.000	29.03±0.000	25.00±0.000	25.27±0.002	24.60±0.000	30.11±0.002	27.02±0.004	28.36±0.013	28.36±0.008	31.85±0.004
	SPamCo	25.00±0.000	25.00±0.000	25.00±0.000	25.00±0.000	25.27±0.002	25.00±0.000	25.00±0.000	25.00±0.000	25.00±0.000	25.00±0.000	25.00±0.000	25.00±0.000
	TMLC	32.88±0.013	35.30±0.033	30.99±0.013	31.94±0.047	36.64±0.003	35.70±0.006	36.51±0.029	32.34±0.003	39.33±0.005	39.19±0.004	38.66±0.024	40.54±0.022
	CDPL	**42.88±0.024**	**49.19±0.029**	**46.64±0.018**	**47.85±0.021**	**48.52±0.031**	**48.79±0.025**	**49.87±0.024**	**47.18±0.011**	**47.85±0.020**	**49.06±0.012**	**50.40±0.008**	**52.96±0.015**
	Δ	**+4.04**	**+8.46**	**+6.45**	**+3.23**	**+5.64**	**+5.38**	**+5.65**	**+1.35**	**+6.86**	**+4.97**	**+4.97**	**+1.21**
Handwritten	CCL	91.08±0.018	91.17±0.019	92.08±0.008	88.00±0.015	94.25±0.007	93.67±0.008	94.00±0.005	86.00±0.012	94.67±0.005	94.42±0.004	93.67±0.008	94.83±0.001
	CPE	50.25±0.149	59.75±0.279	61.00±0.279	46.50±0.516	80.88±0.140	78.38±0.076	49.88±0.507	60.50±0.286	72.00±0.152	82.00±0.149	70.75±0.301	57.62±0.394
	SimPro	81.75±0.013	83.92±0.018	82.08±0.032	78.92±0.089	89.08±0.047	91.42±0.038	90.92±0.039	90.83±0.021	89.00±0.074	89.17±0.040	90.00±0.035	89.17±0.081
	Meta-Expert	91.33±0.008	90.75±0.026	89.50±0.030	91.00±0.023	93.42±0.010	93.33±0.005	92.50±0.004	87.00±0.003	94.25±0.013	95.42±0.004	95.17±0.005	96.50±0.009
	Co-GCN	62.67±0.245	59.38±0.253	58.83±0.280	59.50±0.218	61.67±0.244	59.46±0.242	63.08±0.239	62.08±0.220	66.67±0.174	64.29±0.221	63.75±0.213	66.08±0.187
	DSRL	56.50±0.000	48.50±0.000	51.00±0.000	54.75±0.000	61.25±0.000	58.50±0.000	59.25±0.000	59.50±0.000	67.50±0.000	64.25±0.000	64.25±0.000	64.50±0.000
	GEGCN	78.00±0.208	85.92±0.086	67.08±0.339	87.92±0.073	64.25±0.282	82.75±0.138	92.25±0.049	67.50±0.280	79.33±0.146	92.67±0.056	89.83±0.116	70.92±0.110
	IMvGCN	80.33±0.016	79.00±0.005	80.83±0.023	82.17±0.006	83.33±0.013	81.00±0.013	85.83±0.024	86.50±0.005	91.17±0.003	88.83±0.015	90.67±0.003	91.67±0.013
	MMatch	86.12±0.012	84.75±0.028	86.75±0.032	88.62±0.009	93.75±0.011	92.00±0.014	93.75±0.000	94.75±0.004	92.88±0.002	92.88±0.002	93.12±0.009	94.50±0.004
	SPamCo	21.67±0.001	21.67±0.001	21.67±0.001	21.75±0.003	19.08±0.027	20.17±0.009	20.25±0.008	29.83±0.017	34.08±0.010	34.08±0.010	34.08±0.010	41.42±0.008
	TMLC	83.92±0.017	84.00±0.020	83.75±0.003	82.83±0.007	87.17±0.000	87.25±0.008	87.33±0.002	82.25±0.005	88.00±0.009	88.17±0.002	88.67±0.001	90.33±0.001
	CDPL	**92.42±0.010**	**94.33±0.005**	**93.50±0.007**	**95.25±0.013**	**95.58±0.003**	**95.00±0.007**	**96.08±0.003**	**96.42±0.001**	**95.50±0.004**	**95.58±0.014**	**96.33±0.005**	**96.58±0.005**
	Δ	**+1.09**	**+3.16**	**+1.42**	**+4.25**	**+1.33**	**+1.33**	**+2.08**	**+1.67**	**+0.83**	**+0.16**	**+1.16**	**+0.08**
NoisyMNIST	CCL	80.83±0.003	81.31±0.003	81.37±0.005	81.53±0.004	87.71±0.001	87.95±0.002	87.54±0.005	87.50±0.002	89.75±0.004	90.16±0.004	90.22±0.002	89.69±0.003
	CPE	66.97±0.017	68.57±0.012	65.79±0.013	68.73±0.006	74.46±0.002	78.18±0.004	75.76±0.016	75.74±0.014	79.50±0.004	80.54±0.014	81.43±0.033	79.67±0.006
	SimPro	86.17±0.051	89.46±0.025	89.64±0.007	90.00±0.012	89.59±0.003	92.44±0.030	89.62±0.008	94.90±0.002	94.66±0.021	95.76±0.005	94.36±0.015	93.28±0.051
	Meta-Expert	75.03±0.126	82.04±0.104	82.78±0.074	81.45±0.138	77.63±0.236	89.37±0.016	87.70±0.032	84.46±0.137	91.13±0.017	87.73±0.061	80.08±0.194	91.53±0.013
	Co-GCN	25.32±0.140	28.15±0.092	26.30±0.127	27.96±0.112	30.19±0.156	29.43±0.137	29.14±0.123	30.11±0.120	35.77±0.188	33.02±0.169	32.78±0.163	33.26±0.198
	IMvGCN	80.43±0.015	81.06±0.011	78.47±0.020	81.31±0.012	84.35±0.002	83.99±0.022	83.72±0.015	86.14±0.008	87.59±0.006	87.37±0.005	85.86±0.018	87.14±0.010
	MMatch	89.46±0.003	89.43±0.006	89.83±0.001	90.43±0.005	91.78±0.000	92.50±0.001	92.37±0.000	92.49±0.001	93.53±0.002	93.26±0.002	93.17±0.003	93.50±0.003
	SPamCo	70.02±0.006	61.15±0.011	64.43±0.018	87.53±0.003	82.48±0.083	80.83±0.010	85.66±0.019	81.83±0.027	95.15±0.010	94.98±0.009	95.42±0.005	94.50±0.011
	TMLC	67.85±0.001	67.84±0.004	68.06±0.000	68.60±0.003	72.37±0.000	72.60±0.001	72.39±0.001	72.40±0.000	75.14±0.000	74.88±0.000	75.39±0.000	74.41±0.001
	CDPL	**97.34±0.001**	**97.44±0.002**	**96.73±0.002**	**98.75±0.000**	**98.19±0.001**	**98.04±0.000**	**98.05±0.000**	**98.78±0.001**	**98.28±0.002**	**98.34±0.001**	**98.26±0.001**	**98.78±0.000**
	Δ	**+7.88**	**+7.98**	**+6.90**	**+8.32**	**+6.41**	**+5.54**	**+5.68**	**+3.88**	**+3.13**	**+2.58**	**+2.84**	**+4.28**
OutScene	CCL	63.94±0.004	63.86±0.006	64.18±0.011	60.26±0.031	69.55±0.022	70.35±0.016	70.19±0.005	64.42±0.006	72.28±0.007	73.96±0.016	74.36±0.005	70.99±0.006
	CPE	35.34±0.202	44.87±0.178	43.91±0.179	38.62±0.135	45.99±0.289	63.22±0.029	63.46±0.035	49.04±0.218	47.60±0.304	47.60±0.296	49.28±0.301	43.99±0.267
	SimPro	56.41±0.068	56.65±0.066	53.04±0.010	45.27±0.021	70.99±0.019	64.98±0.040	66.27±0.014	62.02±0.092	64.98±0.132	67.31±0.084	65.63±0.067	64.02±0.057
	Meta-Expert	70.19±0.021	66.75±0.010	68.83±0.020	70.59±0.041	74.92±0.028	71.71±0.019	73.08±0.027	73.72±0.019	73.00±0.004	74.52±0.011	71.07±0.017	76.68±0.017
	Co-GCN	20.31±0.014	20.15±0.023	19.71±0.023	19.71±0.022	21.35±0.014	21.79±0.016	21.51±0.018	20.79±0.043	25.84±0.046	28.25±0.076	24.64±0.039	23.12±0.042
	DSRL	25.00±0.000	22.84±0.000	24.28±0.000		24.76±0.000	26.20±0.000	26.68±0.000		25.24±0.000	26.44±0.000	25.48±0.000	
	GEGCN	67.31±0.074	65.14±0.038	66.35±0.035	57.61±0.022	72.12±0.052	73.88±0.016	74.36±0.006	73.24±0.005	72.92±0.017	69.47±0.035	69.31±0.067	75.72±0.017
	IMvGCN	70.67±0.013	68.59±0.007	70.67±0.035	55.45±0.044	71.15±0.029	66.67±0.045	70.35±0.027	61.86±0.029	71.15±0.017	71.31±0.007	71.96±0.023	68.75±0.005
	MMatch	70.11±0.012	70.59±0.009	69.71±0.015	64.90±0.008	73.88±0.001	72.60±0.008	75.88±0.016	72.28±0.011	72.04±0.005	72.68±0.014	73.88±0.005	76.20±0.004
	SPamCo	12.50±0.000	13.70±0.000	14.26±0.001	12.50±0.000	12.58±0.001	14.26±0.001	12.98±0.000	12.50±0.000	14.26±0.010	13.86±0.015	14.10±0.011	12.58±0.001
	TMLC	62.34±0.012	62.74±0.017	62.98±0.001	53.13±0.012	63.62±0.004	63.94±0.003	63.22±0.023	62.58±0.001	65.46±0.001	66.35±0.007	66.43±0.002	68.59±0.009
	CDPL	**73.72±0.043**	**76.28±0.038**	**75.56±0.027**	**81.01±0.015**	**79.73±0.019**	**80.77±0.013**	**80.13±0.007**	**82.21±0.013**	**78.04±0.012**	**80.21±0.008**	**79.73±0.006**	**83.25±0.010**
	Δ	**+3.05**	**+5.69**	**+4.89**	**+10.42**	**+4.81**	**+6.89**	**+4.25**	**+8.49**	**+5.04**	**+5.69**	**+5.37**	**+6.57**
YouTubeFace20	CCL	96.73±0.002	96.96±0.002	96.77±0.006	96.35±0.007	98.01±0.004	98.34±0.002	98.28±0.001	98.12±0.002	98.70±0.000	98.66±0.002	98.73±0.002	98.54±0.002
	CPE	96.76±0.004	97.41±0.005	96.02±0.022	97.35±0.003	98.45±0.002	98.57±0.002	98.62±0.002	98.56±0.002	99.00±0.002	96.41±0.045	98.89±0.001	98.71±0.003
	SimPro	94.70±0.008	92.33±0.018	91.50±0.027	91.01±0.007	96.09±0.008	96.33±0.007	96.29±0.011	92.83±0.012	95.83±0.005	96.08±0.009	95.63±0.003	93.87±0.012
	Meta-Expert	97.71±0.007	98.38±0.002	97.33±0.012	97.70±0.005	99.44±0.002	99.36±0.004	99.47±0.000	99.40±0.001	99.49±0.001	99.47±0.000	99.52±0.000	99.34±0.003
	Co-GCN	86.15±0.085	85.46±0.080	86.13±0.079	84.88±0.084	90.13±0.072	89.60±0.068	90.11±0.072	88.18±0.079	91.62±0.069	91.12±0.068	91.16±0.067	90.27±0.069
	IMvGCN	98.04±0.006	97.41±0.004	98.50±0.002	98.71±0.003	99.13±0.002	99.47±0.001	99.43±0.000	99.09±0.002	99.37±0.000	99.46±0.000	99.42±0.001	99.25±0.001
	MMatch	97.62±0.000	97.16±0.001	97.25±0.001	97.49±0.003	98.62±0.002	98.66±0.001	98.51±0.001	98.38±0.002	99.31±0.000	99.25±0.001	99.15±0.001	99.06±0.000
	SPamCo	87.89±0.049	84.25±0.024	87.60±0.043	92.65±0.011	94.18±0.013	91.76±0.016	91.87±0.003	93.69±0.009	96.02±0.008	95.27±0.018	94.74±0.016	95.15±0.015
	TMLC	97.29±0.000	97.13±0.000	97.14±0.000	97.61±0.000	97.93±0.000	97.96±0.000	97.95±0.000	98.17±0.000	98.87±0.000	98.84±0.000	98.89±0.000	98.73±0.000
	CDPL	**98.60±0.002**	**99.00±0.000**	**98.83±0.000**	**98.88±0.001**	**99.54±0.000**	**99.55±0.000**	**99.51±0.000**	**99.44±0.002**	**99.51±0.000**	**99.54±0.000**	**99.55±0.000**	**99.37±0.000**
	Δ	**+0.56**	**+0.62**	**+0.33**	**+0.17**	**+0.10**	**+0.08**	**+0.04**	**+0.04**	**+0.02**	**+0.07**	**+0.03**	**+0.03**

Note: C/I/A/U denote consistent, inverse, arbitrary, and uniform distributions. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20. DSRL also ran out of memory for the three OutScene uniform settings; those cells are left empty. Missing runs are excluded from ranking.

TABLE II

G-MEAN (%) ON SEVEN DATASETS AT $LR = 0.09$ AND $IR \in \{0.1, 0.2, 0.3\}$ UNDER FOUR UNLABELED DISTRIBUTIONS. VALUES ARE MEAN ± REPORTED VARIATION. THE BEST AND SECOND-BEST MEANS ARE MARKED BY PINK/BOLD AND BLUE/UNDERLINED ENTRIES, RESPECTIVELY. THE Δ ROW REPORTS CDPL MINUS THE STRONGEST COMPETING METHOD (PERCENTAGE POINTS); POSITIVE/NEGATIVE DIFFERENCES ARE BOLD RED/BLUE.

Dataset	Method	$IR = 0.1$				$IR = 0.2$				$IR = 0.3$			
		C	I	A	U	C	I	A	U	C	I	A	U
CIFAR-100	CCL	72.01±0.004	71.69±0.003	71.40±0.001	72.32±0.004	76.39±0.002	76.19±0.001	76.38±0.004	76.81±0.006	77.62±0.003	77.65±0.003	77.82±0.005	78.05±0.010
	CPE	65.32±0.007	63.85±0.007	65.92±0.002	66.15±0.007	70.36±0.002	70.46±0.008	70.89±0.007	69.24±0.007	72.88±0.006	72.70±0.002	72.75±0.014	73.33±0.003
	SimPro	66.80±0.010	68.72±0.004	68.70±0.013	63.67±0.029	72.65±0.011	73.83±0.006	73.46±0.008	69.66±0.008	74.84±0.010	75.46±0.007	75.52±0.007	72.67±0.012
	Meta-Expert	72.11±0.021	70.37±0.013	72.13±0.011	72.11±0.002	78.19±0.003	77.84±0.003	78.13±0.006	78.56±0.004	79.41±0.001	79.35±0.008	79.61±0.004	80.42±0.003
	Co-GCN	33.29±0.124	34.26±0.119	33.61±0.123	33.72±0.119	35.85±0.145	36.64±0.135	35.84±0.142	36.67±0.128	38.24±0.141	38.24±0.141	38.07±0.142	38.69±0.145
	IMvGCN	63.39±0.003	64.14±0.003	64.00±0.003	65.22±0.002	68.53±0.005	68.59±0.001	68.75±0.002	69.25±0.001	71.36±0.003	71.22±0.001	71.54±0.002	71.87±0.003
	MMatch	20.03±0.001	20.59±0.000	20.76±0.000	21.51±0.004	24.37±0.000	24.47±0.002	24.42±0.006	25.08±0.015	28.14±0.016	27.56±0.005	28.00±0.020	27.84±0.026
	SPamCo	47.21±0.147	56.68±0.003	56.35±0.002	57.02±0.009	63.52±0.003	63.52±0.003	63.52±0.003	64.02±0.002	65.85±0.003	65.85±0.003	65.85±0.003	67.84±0.001
	TMLC	68.45±0.000	68.62±0.001	68.67±0.000	68.29±0.001	72.36±0.000	72.18±0.000	72.43±0.001	72.40±0.003	73.02±0.001	73.15±0.000	73.01±0.000	74.98±0.000
	CDPL	**76.62±0.002**	**76.62±0.002**	**76.62±0.002**	**77.05±0.002**	**79.74±0.002**	**79.74±0.002**	**79.74±0.002**	**79.78±0.003**	**81.10±0.004**	**81.10±0.004**	**81.10±0.004**	**81.26±0.001**
	Δ	**+4.51**	**+4.93**	**+4.49**	**+4.73**	**+1.55**	**+1.90**	**+1.61**	**+1.22**	**+1.69**	**+1.75**	**+1.49**	**+0.84**
CIFAR-10	CCL	62.54±0.017	63.44±0.026	63.31±0.024	63.58±0.016	67.83±0.023	67.77±0.022	68.00±0.024	68.09±0.022	70.31±0.032	70.47±0.033	70.61±0.034	70.24±0.034
	CPE	58.40±0.011	57.19±0.016	57.91±0.007	58.62±0.004	61.89±0.009	61.51±0.003	60.53±0.013	61.50±0.009	63.71±0.007	64.24±0.009	62.75±0.005	64.60±0.012
	SimPro	71.35±0.030	71.61±0.020	71.34±0.022	71.38±0.023	73.84±0.027	73.28±0.023	73.07±0.021	61.68±0.117	74.55±0.018	75.03±0.022	74.43±0.016	70.22±0.069
	Meta-Expert	63.33±0.113	71.75±0.028	64.70±0.110	65.96±0.117	71.80±0.043	74.03±0.019	70.46±0.067	73.23±0.029	74.20±0.040	75.39±0.021	75.09±0.018	73.43±0.046
	Co-GCN	41.99±0.064	44.70±0.038	43.05±0.050	43.29±0.057	42.53±0.067	44.00±0.056	46.03±0.039	44.94±0.041	45.64±0.047	44.15±0.055	44.01±0.058	42.63±0.059
	IMvGCN	65.34±0.002	64.84±0.003	64.81±0.003	65.99±0.001	68.45±0.003	68.17±0.002	68.03±0.001	68.60±0.005	70.42±0.001	69.99±0.002	70.66±0.001	69.88±0.002
	MMatch	38.65±0.006	39.33±0.056	39.53±0.238	51.44±0.027	48.38±0.080	50.05±0.092	52.85±0.198	54.48±0.055	57.28±0.015	55.93±0.022	58.19±0.079	59.35±0.006
	SPamCo	58.00±0.003	57.52±0.002	57.51±0.002	58.41±0.000	63.29±0.004	63.29±0.004	63.29±0.004	64.09±0.001	67.75±0.004	67.59±0.003	67.59±0.003	68.06±0.003
	TMLC	71.00±0.000	70.86±0.000	70.84±0.000	70.22±0.000	72.04±0.000	72.06±0.000	72.01±0.001	71.87±0.000	73.59±0.000	73.40±0.000	73.53±0.001	73.26±0.000
	CDPL	**71.99±0.002**	**72.90±0.000**	**72.26±0.006**	**72.59±0.001**	**75.43±0.004**	**75.29±0.002**	**75.59±0.002**	**75.40±0.003**	**76.99±0.003**	**76.73±0.002**	**76.88±0.001**	**76.73±0.003**
	Δ	**+0.64**	**+1.15**	**+0.92**	**+1.21**	**+1.59**	**+1.26**	**+2.52**	**+2.17**	**+2.44**	**+1.34**	**+1.79**	**+3.30**
GRAZ02	CCL	53.00±0.016	51.73±0.013	50.45±0.005	60.28±0.038	56.50±0.007	54.82±0.009	56.83±0.012	61.26±0.030	56.72±0.004	55.94±0.007	56.94±0.005	65.90±0.006
	CPE	49.06±0.082	46.28±0.042	49.30±0.026	52.13±0.024	49.06±0.082	49.87±0.041	51.10±0.029	49.90±0.093	51.59±0.107	48.04±0.067	47.45±0.080	51.10±0.100
	SimPro	55.59±0.025	56.05±0.019	55.27±0.010	59.02±0.015	56.38±0.021	54.65±0.045	54.24±0.025	55.37±0.021	47.93±0.008	54.26±0.010	53.22±0.020	58.59±0.004
	Meta-Expert	54.23±0.029	53.57±0.011	53.56±0.026	58.92±0.002	57.37±0.027	59.34±0.018	57.43±0.048	55.09±0.047	55.15±0.019	59.89±0.007	60.93±0.034	63.60±0.018
	Co-GCN	47.03±0.031	46.17±0.044	43.68±0.035	45.66±0.031	46.02±0.035	48.11±0.036	47.94±0.032	47.57±0.035	48.73±0.033	47.80±0.012	50.72±0.022	48.94±0.040
	DSRL	47.08±0.000	45.21±0.000	43.30±0.000	52.78±0.000	48.90±0.000	49.62±0.000	43.30±0.000	51.39±0.000	47.81±0.000	43.30±0.000	47.45±0.000	54.49±0.000
	GEGCN	44.68±0.024	44.56±0.021	46.06±0.028	52.30±0.020	48.90±0.010	47.63±0.040	49.19±0.069	54.71±0.013	57.27±0.016	55.77±0.045	55.14±0.027	58.87±0.075
	IMvGCN	55.27±0.009	57.16±0.006	56.72±0.016	57.82±0.015	58.91±0.011	58.37±0.008	60.00±0.011	56.61±0.014	57.38±0.007	57.60±0.018	59.66±0.026	58.59±0.012
	MMatch	0.00±0.000	0.00±0.000	0.00±0.000	19.64±0.006	0.00±0.000	3.34±0.029	0.00±0.000	20.07±0.003	16.91±0.012	19.13±0.036	18.88±0.013	24.89±0.005
	SPamCo	43.30±0.000	43.30±0.000	43.30±0.000	43.30±0.000	43.56±0.002	43.30±0.000	43.30±0.000	43.30±0.000	43.30±0.000	43.30±0.000	43.30±0.000	43.30±0.000
	TMLC	48.91±0.009	50.85±0.021	47.38±0.009	48.14±0.031	51.92±0.002	51.18±0.004	51.81±0.018	48.48±0.002	54.03±0.003	53.92±0.003	53.50±0.015	54.96±0.013
	CDPL	**58.91±0.019**	**63.91±0.023**	**61.92±0.014**	**62.87±0.016**	**63.39±0.024**	**63.60±0.020**	**64.44±0.018**	**62.34±0.008**	**62.87±0.016**	**63.82±0.010**	**64.86±0.006**	**66.82±0.012**
	Δ	**+3.32**	**+6.75**	**+5.20**	**+2.59**	**+4.48**	**+4.26**	**+4.44**	**+1.08**	**+5.49**	**+3.93**	**+3.93**	**+0.92**
Handwritten	CCL	94.96±0.010	95.01±0.011	95.54±0.004	93.18±0.009	96.77±0.004	96.44±0.005	96.63±0.003	92.01±0.007	97.01±0.003	96.87±0.002	96.44±0.004	97.10±0.001
	CPE	68.56±0.108	74.62±0.191	75.49±0.190	60.12±0.426	88.83±0.084	87.42±0.046	63.72±0.398	75.10±0.195	83.33±0.096	89.49±0.089	81.92±0.192	72.04±0.278
	SimPro	89.49±0.008	90.78±0.011	89.68±0.019	87.71±0.055	93.79±0.027	95.14±0.022	94.86±0.023	94.82±0.012	93.72±0.043	93.85±0.023	94.33±0.020	93.80±0.047
	Meta-Expert	95.11±0.004	94.77±0.015	94.04±0.017	94.91±0.013	96.30±0.006	96.25±0.003	95.78±0.003	92.60±0.002	96.77±0.008	97.43±0.002	97.29±0.003	98.04±0.005
	Co-GCN	76.39±0.165	73.98±0.175	73.28±0.195	74.41±0.151	75.66±0.168	74.17±0.167	76.71±0.162	76.22±0.148	79.63±0.113	77.71±0.147	77.40±0.143	79.16±0.122
	DSRL	73.33±0.000	67.62±0.000	69.44±0.000	72.11±0.000	76.56±0.000	74.70±0.000	75.21±0.000	75.38±0.000	80.66±0.000	78.55±0.000	78.55±0.000	78.71±0.000
	GEGCN	86.76±0.130	91.90±0.051	78.59±0.236	93.09±0.042	77.48±0.184	89.91±0.083	95.61±0.028	79.61±0.185	87.84±0.087	95.85±0.032	94.13±0.068	82.69±0.071
	IMvGCN	88.64±0.010	87.84±0.003	88.94±0.014	89.74±0.003	90.44±0.008	89.04±0.008	91.91±0.014	92.30±0.003	95.01±0.002	93.66±0.009	94.72±0.002	95.30±0.007
	MMatch	91.24±0.010	89.66±0.023	91.47±0.025	92.84±0.007	96.43±0.007	95.39±0.009	96.44±0.000	97.04±0.002	95.92±0.001	95.92±0.001	96.05±0.006	96.89±0.002
	SPamCo	44.48±0.002	44.48±0.002	44.48±0.002	44.56±0.003	41.61±0.031	42.86±0.010	42.96±0.008	52.43±0.015	56.20±0.009	56.20±0.009	56.20±0.009	62.22±0.006
	TMLC	90.78±0.006	90.83±0.007	90.68±0.001	90.14±0.002	92.70±0.000	92.74±0.003	92.79±0.001	89.79±0.002	93.18±0.003	93.28±0.001	93.57±0.000	94.53±0.000
	CDPL	**95.73±0.006**	**96.82±0.003**	**96.35±0.004**	**97.34±0.007**	**97.53±0.002**	**97.20±0.004**	**97.81±0.002**	**98.00±0.001**	**97.48±0.002**	**97.53±0.008**	**97.95±0.003**	**98.09±0.003**
	Δ	**+0.62**	**+1.81**	**+0.81**	**+2.43**	**+0.76**	**+0.76**	**+1.18**	**+0.96**	**+0.47**	**+0.10**	**+0.66**	**+0.05**
NoisyMNIST	CCL	88.94±0.002	89.23±0.002	89.27±0.003	89.37±0.002	93.01±0.001	93.15±0.001	92.91±0.003	92.89±0.001	94.20±0.002	94.43±0.003	94.47±0.001	94.16±0.002
	CPE	80.32±0.011	81.35±0.008	79.55±0.009	81.45±0.004	85.06±0.001	87.34±0.002	85.86±0.010	85.84±0.009	88.14±0.003	88.77±0.008	89.29±0.020	88.24±0.004
	SimPro	92.09±0.030	94.02±0.015	94.13±0.004	94.34±0.007	94.10±0.002	95.74±0.017	94.12±0.005	97.14±0.001	97.00±0.012	97.63±0.003	96.83±0.008	96.20±0.029
	Meta-Expert	85.23±0.080	89.57±0.063	90.06±0.045	89.13±0.084	86.37±0.151	93.97±0.009	93.00±0.019	90.93±0.083	94.99±0.010	92.99±0.036	88.10±0.121	95.22±0.007
	Co-GCN	46.54±0.144	50.33±0.088	47.86±0.129	49.89±0.107	51.28±0.146	50.93±0.128	50.87±0.115	51.86±0.110	55.98±0.163	53.80±0.150	53.66±0.146	53.30±0.181
	IMvGCN	88.70±0.009	89.08±0.006	87.52±0.012	89.23±0.007	91.04±0.001	90.82±0.013	90.67±0.009	92.10±0.005	92.94±0.004	92.82±0.003	91.93±0.011	92.68±0.006
	MMatch	93.92±0.002	93.90±0.004	94.16±0.001	94.44±0.003	95.31±0.000	95.74±0.000	95.66±0.000	95.73±0.001	96.34±0.001	96.19±0.001	96.13±0.002	96.32±0.002
	SPamCo	82.27±0.004	76.49±0.007	78.67±0.012	92.91±0.002	89.87±0.050	88.94±0.006	91.81±0.011	89.54±0.016	97.28±0.006	97.18±0.005	97.43±0.003	96.91±0.007
	TMLC	80.88±0.000	80.88±0.002	81.02±0.000	81.37±0.001	83.76±0.000	83.90±0.000	83.76±0.000	83.77±0.000	85.48±0.000	85.32±0.000	85.63±0.000	85.03±0.000
	CDPL	**98.51±0.000**	**98.57±0.001**	**98.17±0.001**	**99.30±0.000**	**98.99±0.000**	**98.91±0.000**	**98.91±0.000**	**99.32±0.001**	**99.04±0.001**	**99.07±0.001**	**99.03±0.001**	**99.32±0.000**
	Δ	**+4.59**	**+4.55**	**+4.01**	**+4.86**	**+3.68**	**+3.17**	**+3.25**	**+2.18**	**+1.76**	**+1.44**	**+1.60**	**+2.41**
OutScene	CCL	77.88±0.003	77.82±0.004	78.04±0.007	75.38±0.022	81.56±0.014	82.08±0.010	81.98±0.003	78.20±0.004	83.31±0.005	84.38±0.010	84.64±0.003	82.49±0.004
	CPE	54.80±0.191	63.42±0.146	62.63±0.148	58.79±0.112	62.45±0.252	77.38±0.020	77.54±0.023	66.26±0.174	63.47±0.264	63.69±0.254	65.04±0.253	61.20±0.236
	SimPro	72.66±0.047	72.83±0.045	70.35±0.007	64.59±0.016	82.49±0.013	78.55±0.027	79.42±0.009	76.47±0.062	78.33±0.090	80.01±0.056	78.94±0.045	77.89±0.038
	Meta-Expert	81.97±0.013	79.73±0.007	81.09±0.013	82.22±0.027	84.98±0.018	82.95±0.013	83.82±0.017	84.23±0.012	83.77±0.002	84.74±0.007	82.54±0.011	86.10±0.011
	Co-GCN	42.41±0.016	42.20±0.025	41.72±0.026	41.72±0.025	43.52±0.015	43.98±0.017	43.67±0.019	42.72±0.050	47.93±0.043	49.99±0.071	46.77±0.039	45.21±0.044
	DSRL	47.25±0.000	45.08±0.000	46.53±0.000		47.01±0.000	48.41±0.000	48.88±0.000		47.48±0.000	48.65±0.000	47.72±0.000	
	GEGCN	80.04±0.049	78.66±0.025	79.46±0.024	73.56±0.015	83.18±0.034	84.33±0.010	84.64±0.004	83.93±0.003	83.72±0.011	81.50±0.023	81.35±0.044	85.49±0.011
	IMvGCN	82.29±0.008	80.94±0.005	82.27±0.022	72.02±0.031	82.59±0.019	79.66±0.030	82.07±0.017	76.47±0.020	82.59±0.011	82.70±0.005	83.10±0.015	81.04±0.003
	MMatch	80.46±0.007	80.90±0.005	80.23±0.008	76.93±0.005	83.70±0.001	82.85±0.007	85.19±0.010	82.35±0.007	82.46±0.003	83.06±0.010	83.86±0.005	85.20±0.003
	SPamCo	33.07±0.000	34.66±0.000	35.38±0.002	33.07±0.000	33.18±0.002	35.38±0.002	33.72±0.000	33.07±0.000	35.36±0.012	34.84±0.019	35.16±0.014	33.18±0.002
	TMLC	76.80±0.005	77.07±0.008	77.23±0.001	70.40±0.006	77.66±0.002	77.88±0.002	77.39±0.011	76.96±0.000	78.89±0.000	79.47±0.003	79.52±0.001	80.94±0.004
	CDPL	**84.21±0.028**	**85.83±0.024**	**85.39±0.017**	**88.77±0.009**	**87.98±0.012**	**88.63±0.008**	**88.23±0.005**	**89.51±0.008**	**86.95±0.008**	**88.28±0.005**	**87.99±0.004**	**90.14±0.006**
	Δ	**+1.92**	**+4.89**	**+3.12**	**+6.55**	**+3.00**	**+4.30**	**+3.04**	**+5.28**	**+3.18**	**+3.54**	**+3.35**	**+4.04**
YouTubeFace20	CCL	98.27±0.001	98.39±0.001	98.29±0.003	98.06±0.004	98.95±0.002	99.13±0.001	99.09±0.001	99.01±0.001	99.31±0.000	99.29±0.001	99.33±0.001	99.23±0.001
	CPE	98.28±0.002	98.63±0.003	97.88±0.012	98.60±0.001	99.18±0.001	99.25±0.001	99.27±0.001	99.24±0.001	99.47±0.001	98.08±0.024	99.42±0.001	99.32±0.001
	SimPro	97.18±0.004	95.89±0.010	95.43±0.015	95.17±0.004	97.92±0.004	98.05±0.004	98.03±0.006	96.16±0.007	97.78±0.003	97.92±0.005	97.68±0.002	96.73±0.006
	Meta-Expert	98.79±0.004	99.15±0.001	98.59±0.006	98.79±0.003	99.71±0.001	99.54±0.004	99.72±0.000	99.68±0.000	99.73±0.000	99.72±0.000	99.75±0.000	99.50±0.004
	Co-GCN	92.39±0.048	92.01±0.045	92.39±0.044	91.68±0.047	94.63±0.039	94.35±0.038	94.62±0.040	93.54±0.044	95.46±0.038	95.18±0.037	95.21±0.037	94.71±0.038
	IMvGCN	98.96±0.003	98.63±0.002	99.21±0.001	99.32±0.001	99.54±0.001	99.72±0.000	99.70±0.000	99.52±0.001	99.67±0.000	99.72±0.000	99.69±0.000	99.61±0.000
	MMatch	98.72±0.000	98.47±0.001	98.52±0.001	98.66±0.001	99.26±0.001	99.28±0.000	99.20±0.001	99.14±0.001	99.63±0.000	99.60±0.001	99.55±0.001	99.50±0.000
	SPamCo	93.43±0.028	91.40±0.013	93.27±0.024	96.07±0.006	96.90±0.007	95.58±0.009	95.64±0.002	96.63±0.005	97.89±0.004	97.48±0.010	97.20±0.009	97.42±0.008
	TMLC	98.56±0.000	98.48±0.000	98.49±0.000	98.73±0.000	98.91±0.000	98.92±0.000	98.92±0.000	99.03±0.000	99.40±0.000	99.39±0.000	99.41±0.000	99.33±0.000
	CDPL	**99.26±0.001**	**99.48±0.000**	**99.38±0.000**	**99.41±0.001**	**99.76±0.000**	**99.76±0.000**	**99.74±0.000**	**99.70±0.001**	**99.74±0.000**	**99.76±0.000**	**99.76±0.000**	**99.67±0.000**
	Δ	**+0.30**	**+0.33**	**+0.17**	**+0.09**	**+0.05**	**+0.04**	**+0.02**	**+0.02**	**+0.01**	**+0.04**	**+0.01**	**+0.06**

Note: C/I/A/U denote consistent, inverse, arbitrary, and uniform distributions. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20. DSRL also ran out of memory for the three OutScene uniform settings; those cells are left empty. Missing runs are excluded from ranking.

TABLE III

SPECIFICITY (%) ON SEVEN DATASETS AT $LR = 0.09$ AND $IR \in \{0.1, 0.2, 0.3\}$ UNDER FOUR UNLABELED DISTRIBUTIONS. VALUES ARE MEAN ± REPORTED VARIATION. THE BEST AND SECOND-BEST MEANS ARE MARKED BY PINK/BOLD AND BLUE/UNDERLINED ENTRIES, RESPECTIVELY. THE Δ ROW REPORTS CDPL MINUS THE STRONGEST COMPETING METHOD (PERCENTAGE POINTS); POSITIVE/NEGATIVE DIFFERENCES ARE BOLD RED/BLUE.

Dataset	Method	$IR = 0.1$				$IR = 0.2$				$IR = 0.3$			
		C	I	A	U	C	I	A	U	C	I	A	U
CIFAR-100	CCL	99.52±0.000	99.51±0.000	99.51±0.000	99.52±0.000	99.58±0.000	99.58±0.000	99.58±0.000	99.59±0.000	99.60±0.000	99.60±0.000	99.60±0.000	99.61±0.000
	CPE	99.42±0.000	99.40±0.000	99.43±0.000	99.43±0.000	99.49±0.000	99.49±0.000	99.50±0.000	99.48±0.000	99.53±0.000	99.53±0.000	99.53±0.000	99.54±0.000
	SimPro	99.44±0.000	99.47±0.000	99.47±0.000	99.40±0.000	99.53±0.000	99.54±0.000	99.54±0.000	99.48±0.000	99.56±0.000	99.57±0.000	99.57±0.000	99.53±0.000
	Meta-Expert	99.52±0.000	99.49±0.000	99.52±0.000	99.52±0.000	99.61±0.000	99.60±0.000	99.61±0.000	99.62±0.000	99.63±0.000	99.63±0.000	99.63±0.000	99.65±0.000
	Co-GCN	99.12±0.001	99.12±0.001	99.12±0.001	99.12±0.001	99.14±0.001	99.14±0.001	99.14±0.001	99.14±0.001	99.16±0.001	99.16±0.001	99.15±0.001	99.16±0.001
	IMvGCN	99.40±0.000	99.41±0.000	99.41±0.000	99.42±0.000	99.47±0.000	99.47±0.000	99.47±0.000	99.48±0.000	99.51±0.000	99.50±0.000	99.51±0.000	99.51±0.000
	MMatch	99.13±0.000	99.13±0.000	99.13±0.000	99.13±0.000	99.15±0.000	99.15±0.000	99.15±0.000	99.15±0.000	99.16±0.000	99.16±0.000	99.16±0.000	99.16±0.000
	SPamCo	99.23±0.001	99.32±0.000	99.31±0.000	99.32±0.000	99.40±0.000	99.40±0.000	99.40±0.000	99.41±0.000	99.43±0.000	99.43±0.000	99.43±0.000	99.46±0.000
	TMLC	99.47±0.000	99.47±0.000	99.47±0.000	99.46±0.000	99.52±0.000	99.52±0.000	99.52±0.000	99.52±0.000	99.53±0.000	99.53±0.000	99.53±0.000	99.56±0.000
	CDPL	**99.59±0.000**	**99.59±0.000**	**99.59±0.000**	**99.59±0.000**	**99.63±0.000**	**99.63±0.000**	**99.63±0.000**	**99.64±0.000**	**99.66±0.000**	**99.66±0.000**	**99.66±0.000**	**99.66±0.000**
	Δ	**+0.07**	**+0.08**	**+0.07**	**+0.07**	**+0.02**	**+0.03**	**+0.02**	**+0.02**	**+0.03**	**+0.03**	**+0.03**	**+0.01**
CIFAR-10	CCL	93.54±0.002	93.67±0.004	93.65±0.003	93.69±0.002	94.31±0.003	94.30±0.003	94.34±0.004	94.35±0.003	94.70±0.005	94.72±0.005	94.74±0.005	94.69±0.005
	CPE	92.97±0.002	92.81±0.002	92.90±0.001	93.00±0.001	93.44±0.001	93.39±0.001	93.26±0.002	93.39±0.001	93.70±0.001	93.78±0.001	93.56±0.001	93.83±0.002
	SimPro	94.86±0.005	94.90±0.003	94.85±0.004	94.86±0.004	95.25±0.004	95.16±0.004	95.13±0.003	93.50±0.016	95.37±0.003	95.44±0.004	95.35±0.003	94.70±0.011
	Meta-Expert	93.73±0.015	94.92±0.004	93.92±0.015	94.11±0.017	94.93±0.007	95.28±0.003	94.74±0.010	95.16±0.005	95.32±0.006	95.50±0.003	95.45±0.003	95.20±0.007
	Co-GCN	91.08±0.006	91.33±0.004	91.17±0.005	91.20±0.006	91.14±0.007	91.27±0.006	91.48±0.004	91.36±0.004	91.44±0.005	91.29±0.006	91.28±0.006	91.14±0.006
	IMvGCN	93.94±0.000	93.87±0.000	93.86±0.000	94.03±0.000	94.40±0.001	94.36±0.000	94.34±0.000	94.43±0.001	94.71±0.000	94.64±0.000	94.74±0.000	94.62±0.000
	MMatch	92.12±0.000	92.21±0.000	92.16±0.001	92.96±0.000	92.75±0.000	92.80±0.000	93.05±0.001	93.14±0.001	93.43±0.000	93.26±0.000	93.48±0.001	93.58±0.000
	SPamCo	92.91±0.000	92.85±0.000	92.85±0.000	92.97±0.000	93.64±0.001	93.64±0.001	93.64±0.001	93.76±0.000	94.30±0.001	94.27±0.001	94.27±0.001	94.35±0.000
	TMLC	94.80±0.000	94.78±0.000	94.77±0.000	94.68±0.000	94.96±0.000	94.96±0.000	94.96±0.000	94.93±0.000	95.21±0.000	95.18±0.000	95.20±0.000	95.16±0.000
	CDPL	**94.95±0.000**	**95.10±0.000**	**95.00±0.001**	**95.05±0.000**	**95.51±0.001**	**95.49±0.000**	**95.53±0.000**	**95.50±0.001**	**95.77±0.001**	**95.72±0.000**	**95.75±0.000**	**95.72±0.001**
	Δ	**+0.09**	**+0.18**	**+0.15**	**+0.19**	**+0.26**	**+0.21**	**+0.40**	**+0.34**	**+0.40**	**+0.22**	**+0.30**	**+0.52**
GRAZ02	CCL	78.58±0.006	78.09±0.005	77.60±0.002	81.54±0.016	79.97±0.003	79.30±0.004	80.11±0.005	81.94±0.013	80.06±0.002	79.75±0.003	80.15±0.002	83.92±0.003
	CPE	77.15±0.030	76.08±0.015	77.17±0.010	78.25±0.009	77.15±0.030	77.40±0.016	77.86±0.011	77.49±0.035	78.15±0.041	76.75±0.025	76.55±0.029	77.95±0.038
	SimPro	79.61±0.010	79.79±0.008	79.48±0.004	81.00±0.006	79.93±0.009	79.26±0.018	79.08±0.010	79.53±0.009	76.66±0.003	79.08±0.004	78.67±0.008	80.82±0.002
	Meta-Expert	79.08±0.011	78.81±0.004	78.81±0.010	80.96±0.001	80.33±0.011	81.14±0.008	80.38±0.019	79.44±0.019	79.44±0.008	81.36±0.003	81.81±0.014	82.93±0.008
	Co-GCN	76.34±0.011	76.05±0.016	75.16±0.012	75.85±0.011	75.99±0.013	76.75±0.013	76.68±0.012	76.55±0.013	76.97±0.012	76.61±0.005	77.71±0.008	77.06±0.015
	DSRL	76.34±0.000	75.67±0.000	75.00±0.000	78.49±0.000	77.02±0.000	77.28±0.000	75.00±0.000	77.96±0.000	76.61±0.000	75.00±0.000	76.48±0.000	79.17±0.000
	GEGCN	75.49±0.009	75.45±0.007	75.99±0.010	78.32±0.008	77.02±0.004	76.57±0.014	77.20±0.025	79.26±0.005	80.29±0.006	79.70±0.018	79.44±0.011	81.00±0.031
	IMvGCN	79.48±0.003	80.24±0.002	80.06±0.006	80.51±0.006	80.96±0.005	80.73±0.003	81.41±0.005	80.02±0.005	80.33±0.003	80.42±0.007	81.27±0.011	80.82±0.005
	MMatch	74.96±0.001	75.00±0.000	75.00±0.000	76.34±0.000	75.00±0.000	75.09±0.001	74.87±0.000	76.70±0.001	75.67±0.001	76.12±0.004	76.12±0.003	77.28±0.001
	SPamCo	75.00±0.000	75.00±0.000	75.00±0.000	75.00±0.000	75.09±0.001	75.00±0.000	75.00±0.000	75.00±0.000	75.00±0.000	75.00±0.000	75.00±0.000	75.00±0.000
	TMLC	**80.96±0.001**	78.77±0.004	77.33±0.001	77.65±0.005	79.21±0.000	78.90±0.001	79.17±0.003	77.78±0.000	80.11±0.001	80.06±0.000	79.89±0.003	80.51±0.002
	CDPL	**80.96±0.008**	**83.06±0.010**	**82.21±0.006**	**82.62±0.007**	**82.84±0.010**	**82.93±0.008**	**83.29±0.008**	**82.39±0.004**	**82.62±0.007**	**83.02±0.004**	**83.47±0.003**	**84.32±0.005**
	Δ	**0.00**	**+2.82**	**+2.15**	**+1.08**	**+1.88**	**+1.79**	**+1.88**	**+0.45**	**+2.29**	**+1.66**	**+1.66**	**+0.40**
Handwritten	CCL	99.01±0.002	99.02±0.002	99.12±0.001	98.67±0.002	99.36±0.001	99.30±0.001	99.33±0.001	98.44±0.001	99.41±0.001	99.38±0.000	99.30±0.001	99.43±0.000
	CPE	94.47±0.017	95.53±0.031	95.67±0.031	94.06±0.057	97.88±0.016	97.60±0.008	94.43±0.056	95.61±0.032	96.89±0.017	98.00±0.017	96.75±0.033	95.29±0.044
	SimPro	97.97±0.001	98.21±0.002	98.01±0.004	97.66±0.010	98.79±0.005	99.05±0.004	98.99±0.004	98.98±0.002	98.78±0.008	98.80±0.004	98.89±0.004	98.80±0.009
	Meta-Expert	99.04±0.001	98.97±0.003	98.83±0.003	99.00±0.003	99.27±0.001	99.26±0.001	99.17±0.001	98.56±0.000	99.36±0.002	99.49±0.000	99.46±0.001	99.61±0.001
	Co-GCN	95.85±0.027	95.49±0.028	95.43±0.031	95.50±0.024	95.74±0.027	95.50±0.027	95.90±0.027	95.79±0.024	96.30±0.019	96.03±0.025	95.97±0.024	96.23±0.021
	DSRL	95.17±0.000	94.28±0.000	94.56±0.000	94.97±0.000	95.69±0.000	95.39±0.000	95.47±0.000	95.50±0.000	96.39±0.000	96.03±0.000	96.03±0.000	96.06±0.000
	GEGCN	97.56±0.023	98.44±0.010	96.34±0.038	98.66±0.008	96.03±0.031	98.08±0.015	99.14±0.006	96.39±0.031	97.70±0.016	99.19±0.006	98.87±0.013	96.77±0.012
	IMvGCN	97.81±0.002	97.67±0.001	97.87±0.003	98.02±0.001	98.15±0.001	97.89±0.002	98.43±0.003	98.50±0.001	99.02±0.000	98.76±0.002	98.96±0.000	99.07±0.001
	MMatch	98.46±0.001	98.31±0.003	98.53±0.004	98.74±0.001	99.31±0.001	99.11±0.002	99.31±0.000	99.42±0.000	99.21±0.000	99.21±0.000	99.24±0.001	99.39±0.000
	SPamCo	91.30±0.000	91.30±0.000	91.30±0.000	91.31±0.000	91.01±0.003	91.13±0.001	91.14±0.001	92.20±0.002	92.68±0.001	92.68±0.001	92.68±0.001	93.49±0.001
	TMLC	98.21±0.000	98.22±0.000	98.19±0.000	98.09±0.000	98.57±0.000	98.58±0.000	98.59±0.000	98.03±0.000	98.67±0.000	98.69±0.000	98.74±0.000	98.93±0.000
	CDPL	**99.16±0.001**	**99.37±0.001**	**99.28±0.001**	**99.47±0.001**	**99.51±0.000**	**99.44±0.001**	**99.56±0.000**	**99.60±0.000**	**99.50±0.001**	**99.51±0.002**	**99.59±0.001**	**99.62±0.001**
	Δ	**+0.12**	**+0.35**	**+0.16**	**+0.47**	**+0.15**	**+0.14**	**+0.23**	**+0.18**	**+0.09**	**+0.02**	**+0.13**	**+0.01**
NoisyMNIST	CCL	97.87±0.000	97.92±0.000	97.93±0.001	97.95±0.001	98.63±0.000	98.66±0.000	98.62±0.001	98.61±0.000	98.86±0.000	98.91±0.001	98.91±0.000	98.85±0.000
	CPE	96.33±0.002	96.51±0.001	96.20±0.002	96.53±0.001	97.16±0.000	97.58±0.000	97.31±0.002	97.30±0.002	97.72±0.001	97.84±0.002	97.94±0.004	97.74±0.001
	SimPro	98.46±0.006	98.83±0.003	98.85±0.001	98.89±0.001	98.84±0.000	99.16±0.003	98.85±0.001	99.43±0.000	99.41±0.002	99.53±0.001	99.37±0.002	99.25±0.006
	Meta-Expert	97.23±0.014	98.00±0.012	98.09±0.008	97.94±0.015	97.51±0.026	98.82±0.002	98.63±0.004	98.27±0.015	99.01±0.002	98.64±0.007	97.79±0.022	99.06±0.001
	Co-GCN	91.70±0.016	92.02±0.010	91.81±0.014	92.00±0.013	92.24±0.017	92.16±0.015	92.13±0.014	92.23±0.013	92.86±0.021	92.56±0.019	92.53±0.018	92.58±0.022
	IMvGCN	97.83±0.002	97.90±0.001	97.61±0.002	97.92±0.001	98.26±0.000	98.22±0.002	98.19±0.002	98.46±0.001	98.62±0.001	98.60±0.001	98.43±0.002	98.57±0.001
	MMatch	98.83±0.000	98.83±0.001	98.87±0.000	98.94±0.001	99.09±0.000	99.17±0.000	99.15±0.000	99.17±0.000	99.28±0.000	99.25±0.000	99.24±0.000	99.28±0.000
	SPamCo	96.67±0.001	95.68±0.001	96.05±0.002	98.61±0.000	98.05±0.009	97.87±0.001	98.41±0.002	97.98±0.003	99.46±0.001	99.44±0.001	99.49±0.001	99.39±0.001
	TMLC	96.43±0.000	96.43±0.000	96.45±0.000	96.51±0.000	96.93±0.000	96.96±0.000	96.93±0.000	96.93±0.000	97.24±0.000	97.21±0.000	97.27±0.000	97.16±0.000
	CDPL	**99.70±0.000**	**99.72±0.000**	**99.64±0.000**	**99.86±0.000**	**99.80±0.000**	**99.78±0.000**	**99.78±0.000**	**99.86±0.000**	**99.81±0.000**	**99.82±0.000**	**99.81±0.000**	**99.86±0.000**
	Δ	**+0.87**	**+0.89**	**+0.77**	**+0.92**	**+0.71**	**+0.61**	**+0.63**	**+0.43**	**+0.35**	**+0.29**	**+0.32**	**+0.47**
OutScene	CCL	94.85±0.001	94.84±0.001	94.88±0.002	94.32±0.005	95.65±0.003	95.76±0.002	95.74±0.001	94.92±0.001	96.04±0.001	96.28±0.002	96.34±0.001	95.86±0.001
	CPE	90.76±0.029	92.12±0.026	91.99±0.026	91.23±0.019	92.28±0.041	94.75±0.004	94.78±0.005	92.72±0.031	92.51±0.044	92.51±0.042	92.75±0.043	92.00±0.038
	SimPro	93.77±0.010	93.81±0.009	93.29±0.001	92.18±0.003	95.86±0.003	95.00±0.006	95.18±0.002	94.57±0.013	95.00±0.019	95.33±0.012	95.09±0.010	94.86±0.008
	Meta-Expert	95.74±0.003	95.25±0.001	95.55±0.003	95.80±0.006	96.42±0.004	95.96±0.003	96.15±0.004	96.25±0.003	96.14±0.001	96.36±0.002	95.87±0.002	96.67±0.002
	Co-GCN	88.62±0.002	88.59±0.003	88.53±0.003	88.53±0.003	88.76±0.002	88.83±0.002	88.79±0.003	88.68±0.006	89.41±0.007	89.75±0.011	89.23±0.006	89.02±0.006
	DSRL	89.29±0.000	88.98±0.000	89.18±0.000		89.25±0.000	89.46±0.000	89.53±0.000		89.32±0.000	89.49±0.000	89.35±0.000	
	GEGCN	95.33±0.011	95.02±0.005	95.19±0.005	93.94±0.003	96.02±0.007	96.27±0.002	96.34±0.001	96.18±0.001	96.13±0.003	95.64±0.005	95.62±0.010	96.53±0.003
	IMvGCN	95.81±0.002	95.51±0.001	95.81±0.005	93.64±0.006	95.88±0.004	95.24±0.007	95.76±0.004	94.55±0.004	95.88±0.003	95.90±0.001	95.99±0.003	95.54±0.001
	MMatch	95.73±0.002	95.80±0.001	95.67±0.002	94.99±0.001	96.27±0.000	96.09±0.001	96.55±0.002	96.04±0.002	96.01±0.001	96.10±0.002	96.27±0.001	96.60±0.001
	SPamCo	87.50±0.000	87.67±0.000	87.75±0.000	87.50±0.000	87.51±0.000	87.75±0.000	87.57±0.000	87.50±0.000	87.75±0.001	87.69±0.002	87.73±0.002	87.51±0.000
	TMLC	94.62±0.000	94.68±0.000	94.71±0.000	93.30±0.000	94.80±0.000	94.85±0.000	94.75±0.000	94.65±0.000	95.07±0.000	95.19±0.000	95.20±0.000	95.51±0.000
	CDPL	**96.25±0.006**	**96.61±0.005**	**96.51±0.004**	**97.29±0.002**	**97.10±0.003**	**97.25±0.002**	**97.16±0.001**	**97.46±0.002**	**96.86±0.002**	**97.17±0.001**	**97.10±0.001**	**97.61±0.001**
	Δ	**+0.44**	**+0.81**	**+0.70**	**+1.49**	**+0.68**	**+0.98**	**+0.61**	**+1.21**	**+0.72**	**+0.81**	**+0.76**	**+0.94**
YouTubeFace20	CCL	99.83±0.000	99.84±0.000	99.83±0.000	99.81±0.000	99.90±0.000	99.91±0.000	99.91±0.000	99.90±0.000	99.93±0.000	99.93±0.000	99.93±0.000	99.92±0.000
	CPE	99.83±0.000	99.86±0.000	99.79±0.001	99.86±0.000	99.92±0.000	99.92±0.000	99.93±0.000	99.92±0.000	99.95±0.000	99.81±0.002	99.94±0.000	99.93±0.000
	SimPro	99.72±0.000	99.60±0.001	99.55±0.001	99.53±0.000	99.79±0.000	99.81±0.000	99.80±0.001	99.62±0.001	99.78±0.000	99.79±0.001	99.77±0.000	99.68±0.001
	Meta-Expert	99.88±0.000	99.91±0.000	99.86±0.001	99.88±0.000	99.97±0.000	99.68±0.005	**99.97±0.000**	**99.97±0.000**	**99.97±0.000**	99.97±0.000	99.97±0.000	99.67±0.005
	Co-GCN	99.27±0.005	99.23±0.004	99.27±0.004	99.20±0.004	99.48±0.004	99.45±0.004	99.48±0.004	99.38±0.004	99.56±0.004	99.53±0.004	99.53±0.004	99.49±0.004
	IMvGCN	99.90±0.000	99.86±0.000	99.92±0.000	99.93±0.000	99.95±0.000	99.97±0.000	**99.97±0.000**	99.95±0.000	**99.97±0.000**	99.97±0.000	99.97±0.000	99.96±0.000
	MMatch	99.87±0.000	99.85±0.000	99.86±0.000	99.87±0.000	99.93±0.000	99.93±0.000	99.92±0.000	99.91±0.000	99.96±0.000	99.96±0.000	99.96±0.000	99.95±0.000
	SPamCo	99.36±0.003	99.17±0.001	99.35±0.002	99.61±0.001	99.69±0.001	99.57±0.001	99.57±0.000	99.67±0.001	99.79±0.000	99.75±0.001	99.72±0.001	99.74±0.001
	TMLC	99.86±0.000	99.85±0.000	99.85±0.000	99.87±0.000	99.89±0.000	99.89±0.000	99.89±0.000	99.90±0.000	99.94±0.000	99.94±0.000	99.94±0.000	99.93±0.000
	CDPL	**99.93±0.000**	**99.95±0.000**	**99.94±0.000**	**99.94±0.000**	**99.98±0.000**	**99.98±0.000**	**99.97±0.000**	**99.97±0.000**	**99.97±0.000**	**99.98±0.000**	**99.98±0.000**	**99.97±0.000**
	Δ	**+0.03**	**+0.04**	**+0.02**	**+0.01**	**+0.01**	**+0.01**	**0.00**	**0.00**	**0.00**	**+0.01**	**+0.01**	**+0.01**

Note: C/I/A/U denote consistent, inverse, arbitrary, and uniform distributions. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20. DSRL also ran out of memory for the three OutScene uniform settings; those cells are left empty. Missing runs are excluded from ranking.

TABLE IV

F1 SCORES (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

Dataset	Method	LR=0.03	LR=0.06	LR=0.09	LR=0.12	LR=0.15
CIFAR100	CCL	30.59	45.12	50.46	47.66	48.08
	CPE	21.35	35.71	40.35	38.73	38.72
	SimPro	27.09	37.63	44.58	45.42	44.39
	Meta-Expert	24.44	42.84	53.38	50.79	50.12
	Co-GCN	12.62	14.16	8.12	14.72	14.53
	DSRL	-	-	-	-	-
	GEGCN	-	-	-	-	-
	IMvGCN	28.88	35.19	36.82	39.58	40.42
	MMatch	8.78	8.14	7.00	7.33	6.98
	SPamCo	13.73	20.40	20.42	32.06	36.84
	TMLC	34.32	38.57	39.43	40.72	41.25
	Ours	**42.31**	**53.86**	**58.27**	**62.20**	**64.29**
	Δ	**+7.99**	**+8.74**	**+4.89**	**+11.41**	**+14.17**
GRAZ02	CCL	34.81	33.14	32.15	28.63	30.43
	CPE	31.74	23.19	21.65	27.03	27.75
	SimPro	35.96	27.34	36.08	25.79	37.54
	Meta-Expert	23.12	32.15	36.20	32.63	35.15
	Co-GCN	13.91	15.23	18.36	12.73	16.68
	DSRL	20.70	10.00	23.21	10.00	10.00
	GEGCN	17.14	21.91	13.40	20.69	28.10
	IMvGCN	34.26	29.52	38.13	32.89	40.46
	MMatch	12.12	10.00	9.96	10.83	13.67
	SPamCo	10.00	10.00	10.00	10.55	10.00
	TMLC	34.41	30.99	33.04	26.44	38.10
	Ours	**39.80**	**40.01**	**40.89**	**39.12**	**49.12**
	Δ	**+3.84**	**+6.87**	**+2.76**	**+6.23**	**+8.66**
NoisyMNIST	CCL	68.22	77.05	80.23	80.69	81.53
	CPE	46.61	60.33	65.32	65.79	61.51
	SimPro	79.02	82.87	86.20	90.41	84.96
	Meta-Expert	77.31	81.50	75.75	86.47	84.94
	Co-GCN	22.86	22.58	14.06	22.93	23.29
	IMvGCN	72.69	77.64	79.59	80.35	79.31
	MMatch	81.54	88.02	89.35	90.67	90.83
	SPamCo	26.10	40.88	63.02	64.83	80.69
	TMLC	61.24	65.27	65.75	67.47	69.03
	Ours	**96.95**	**97.44**	**97.34**	**97.45**	**97.63**
	Δ	**+15.41**	**+9.42**	**+7.99**	**+6.78**	**+6.80**
OutScene	CCL	49.57	64.18	62.13	59.18	55.79
	CPE	36.80	43.63	29.91	45.17	41.77
	SimPro	41.85	48.47	55.11	52.82	48.76
	Meta-Expert	50.79	66.73	67.66	66.90	59.54
	Co-GCN	14.49	15.79	13.00	13.11	12.66
	DSRL	22.53	25.22	23.18	24.86	21.96
	GEGCN	36.71	48.67	62.88	73.91	67.99
	IMvGCN	57.84	66.00	69.37	69.28	64.83
	MMatch	53.53	63.32	68.44	69.34	70.56
	SPamCo	2.78	2.78	2.78	2.78	5.58
	TMLC	44.38	54.28	60.88	63.52	58.44
	Ours	**60.95**	**68.24**	**73.36**	**76.99**	**74.39**
	Δ	**+3.11**	**+1.51**	**+3.99**	**+3.08**	**+3.83**

TABLE V

ACCURACY (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

Dataset	Method	LR=0.03	LR=0.06	LR=0.09	LR=0.12	LR=0.15
CIFAR100	CCL	33.30	47.33	52.11	50.05	50.11
	CPE	23.79	38.38	42.91	41.99	41.45
	SimPro	26.84	38.04	44.87	45.99	45.11
	Meta-Expert	21.98	39.53	52.29	49.22	50.23
	Co-GCN	19.03	20.51	12.47	21.25	21.49
	DSRL	-	-	-	-	-
	GEGCN	-	-	-	-	-
	IMvGCN	32.95	39.72	40.43	42.65	43.98
	MMatch	13.72	14.13	13.72	13.79	13.56
	SPamCo	18.26	25.10	23.89	35.38	40.45
	TMLC	40.88	45.93	47.11	48.63	49.06
	Ours	**43.24**	**54.38**	**58.94**	**62.87**	**64.86**
	Δ	**+2.36**	**+7.05**	**+6.65**	**+12.82**	**+14.63**
GRAZ02	CCL	38.04	36.56	35.75	33.74	33.06
	CPE	35.08	29.97	31.45	32.80	34.01
	SimPro	38.71	33.20	38.84	30.51	41.94
	Meta-Expert	32.12	35.35	37.23	37.23	35.22
	Co-GCN	22.98	26.48	29.03	23.39	28.63
	DSRL	25.00	25.00	29.03	25.00	25.00
	GEGCN	24.33	31.05	26.48	28.63	35.62
	IMvGCN	37.23	30.78	38.44	34.68	41.13
	MMatch	26.08	25.00	24.87	24.87	26.75
	SPamCo	25.00	25.00	25.00	25.27	25.00
	TMLC	34.35	31.40	32.88	28.04	37.98
	Ours	**40.73**	**40.05**	**42.88**	**41.13**	**50.00**
	Δ	**+2.02**	**+3.49**	**+4.04**	**+3.90**	**+8.06**
NoisyMNIST	CCL	70.53	77.98	80.83	81.33	81.97
	CPE	50.32	61.90	66.97	67.84	63.65
	SimPro	79.74	82.69	86.17	90.39	84.45
	Meta-Expert	79.62	81.12	75.03	86.71	84.82
	Co-GCN	38.25	38.07	25.32	38.54	38.28
	IMvGCN	74.43	78.94	80.43	81.27	80.87
	MMatch	82.28	88.23	89.46	90.81	90.97
	SPamCo	38.34	50.01	70.02	70.82	83.42
	TMLC	63.82	67.42	67.85	69.48	71.09
	Ours	**96.95**	**97.44**	**97.34**	**97.45**	**97.63**
	Δ	**+14.67**	**+9.21**	**+7.88**	**+6.64**	**+6.66**
OutScene	CCL	54.86	65.46	63.94	61.06	59.21
	CPE	42.55	46.63	35.34	47.68	46.23
	SimPro	44.87	48.56	56.41	53.77	50.56
	Meta-Expert	55.61	67.63	70.19	68.19	61.38
	Co-GCN	20.19	21.79	20.31	19.47	20.75
	DSRL	24.76	25.96	25.00	26.92	23.80
	GEGCN	44.79	54.17	67.31	74.12	69.95
	IMvGCN	**61.78**	67.15	70.67	70.91	67.95
	MMatch	53.85	65.71	70.11	70.67	72.36
	SPamCo	12.50	12.50	12.50	12.50	13.94
	TMLC	49.12	56.97	62.34	65.38	60.18
	Ours	61.70	**67.71**	**73.72**	**76.84**	**74.36**
	Δ	**-0.08**	**+0.08**	**+3.05**	**+2.72**	**+2.00**

TABLE VI

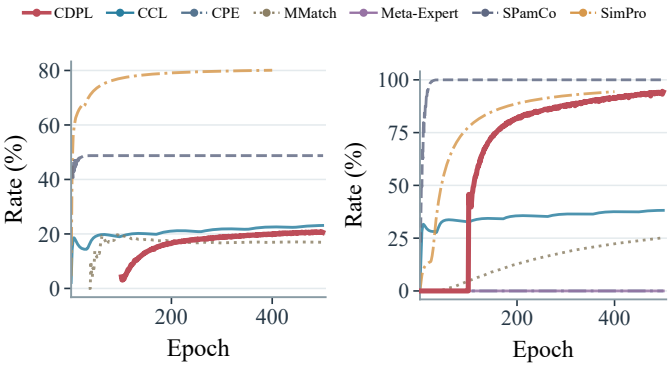
G-MEAN (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

Dataset	Method	LR=0.03	LR=0.06	LR=0.09	LR=0.12	LR=0.15
CIFAR100	CCL	57.51	68.61	72.01	70.57	70.61
	CPE	48.59	61.75	65.32	64.61	64.18
	SimPro	51.55	61.47	66.80	67.63	66.97
	Meta-Expert	46.56	62.61	72.11	69.95	70.68
	Co-GCN	43.44	45.10	33.29	45.92	46.18
	DSRL	-	-	-	-	-
	GEGCN	-	-	-	-	-
	IMvGCN	57.21	62.83	63.39	65.12	66.13
	MMatch	24.51	22.81	20.03	20.58	19.93
	SPamCo	42.55	49.91	47.21	59.29	63.41
	TMLC	63.75	67.59	68.45	69.55	69.86
	Ours	**65.57**	**73.57**	**76.62**	**79.14**	**80.39**
	Δ	**+1.82**	**+4.96**	**+4.51**	**+8.57**	**+9.71**
GRAZ02	CCL	54.94	53.68	53.00	51.26	50.68
	CPE	52.42	47.90	49.06	50.44	51.50
	SimPro	55.49	50.78	55.59	48.41	58.15
	Meta-Expert	49.82	52.58	54.23	54.25	52.45
	Co-GCN	41.22	44.66	47.03	41.55	46.69
	DSRL	43.30	43.30	47.08	43.30	43.30
	GEGCN	42.63	48.89	44.68	46.71	52.85
	IMvGCN	54.26	48.62	55.27	52.07	57.49
	MMatch	7.76	0.00	0.00	4.52	10.84
	SPamCo	43.30	43.30	43.30	43.56	43.30
	TMLC	50.10	47.71	48.91	44.93	52.97
	Ours	**57.13**	**56.61**	**58.91**	**57.48**	**64.55**
	Δ	**+1.64**	**+2.93**	**+3.32**	**+3.23**	**+6.40**
NoisyMNIST	CCL	82.59	87.22	88.94	89.24	89.62
	CPE	68.94	76.98	80.32	80.87	78.14
	SimPro	88.28	90.05	92.09	94.55	91.07
	Meta-Expert	88.22	89.12	85.23	92.42	91.32
	Co-GCN	59.69	59.54	46.54	59.93	59.71
	IMvGCN	85.04	87.80	88.70	89.20	88.97
	MMatch	89.18	93.14	93.92	94.70	94.79
	SPamCo	59.76	68.72	82.27	82.78	90.49
	TMLC	78.27	80.61	80.88	81.93	82.95
	Ours	**98.29**	**98.57**	**98.51**	**98.58**	**98.68**
	Δ	**+9.11**	**+5.43**	**+4.59**	**+3.88**	**+3.89**
OutScene	CCL	70.11	78.89	77.88	75.93	74.68
	CPE	62.37	65.62	54.80	66.37	65.32
	SimPro	64.28	67.03	72.66	70.77	68.40
	Meta-Expert	72.16	79.60	81.97	80.66	76.15
	Co-GCN	42.29	44.00	42.41	41.51	42.89
	DSRL	47.01	48.18	47.25	49.10	46.05
	GEGCN	64.13	71.14	80.04	84.49	81.82
	IMvGCN	**76.42**	80.00	82.29	82.44	80.52
	MMatch	69.11	77.02	80.46	81.04	81.65
	SPamCo	33.07	33.07	33.07	33.07	34.97
	TMLC	67.47	73.12	76.80	78.84	75.33
	Ours	76.37	**80.35**	**84.21**	**86.19**	**84.64**
	Δ	**-0.05**	**+0.35**	**+1.92**	**+1.70**	**+2.82**

TABLE VII

SPECIFICITY (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

Dataset	Method	LR=0.03	LR=0.06	LR=0.09	LR=0.12	LR=0.15
CIFAR100	CCL	99.33	99.47	99.52	99.50	99.50
	CPE	99.23	99.38	99.42	99.41	99.41
	SimPro	99.26	99.37	99.44	99.45	99.45
	Meta-Expert	99.21	99.39	99.52	99.49	99.50
	Co-GCN	99.18	99.20	99.12	99.20	99.21
	DSRL	-	-	-	-	-
	GEGCN	-	-	-	-	-
	IMvGCN	99.32	99.39	99.40	99.42	99.43
	MMatch	99.13	99.13	99.13	99.13	99.13
	SPamCo	99.17	99.24	99.23	99.35	99.40
	TMLC	99.40	99.45	99.47	99.48	99.49
	Ours	**99.43**	**99.54**	**99.59**	**99.62**	**99.65**
	Δ	**+0.03**	**+0.07**	**+0.07**	**+0.12**	**+0.15**
GRAZ02	CCL	79.35	78.85	78.58	77.91	77.69
	CPE	78.36	76.66	77.15	77.60	78.00
	SimPro	79.57	77.73	79.61	76.84	80.65
	Meta-Expert	77.37	78.45	79.08	79.08	78.41
	Co-GCN	74.33	75.49	76.34	74.46	76.21
	DSRL	75.00	75.00	76.34	75.00	75.00
	GEGCN	74.78	77.02	75.49	76.21	78.54
	IMvGCN	79.08	76.93	79.48	78.23	80.38
	MMatch	75.36	75.00	74.96	74.96	75.58
	SPamCo	75.00	75.00	75.00	75.09	75.00
	TMLC	78.45	77.47	77.96	76.35	79.66
	Ours	**80.24**	**80.02**	**80.96**	**80.38**	**83.33**
	Δ	**+0.67**	**+1.17**	**+1.35**	**+1.30**	**+2.68**
NoisyMNIST	CCL	96.73	97.55	97.87	97.93	98.00
	CPE	94.48	95.77	96.33	96.43	95.96
	SimPro	97.75	98.08	98.46	98.93	98.27
	Meta-Expert	97.74	97.90	97.23	98.52	98.31
	Co-GCN	93.14	93.12	91.70	93.17	93.14
	IMvGCN	97.16	97.66	97.83	97.92	97.87
	MMatch	98.03	98.69	98.83	98.98	99.00
	SPamCo	93.15	94.45	96.67	96.76	98.16
	TMLC	95.98	96.38	96.43	96.61	96.79
	Ours	**99.66**	**99.72**	**99.70**	**99.72**	**99.74**
	Δ	**+1.63**	**+1.03**	**+0.87**	**+0.74**	**+0.74**
OutScene	CCL	93.47	95.07	94.85	94.44	94.17
	CPE	91.79	92.38	90.76	92.53	92.32
	SimPro	92.12	92.65	93.77	93.40	92.94
	Meta-Expert	93.66	95.36	95.74	95.46	94.48
	Co-GCN	88.60	88.83	88.62	88.50	88.68
	DSRL	89.25	89.42	89.29	89.56	89.11
	GEGCN	92.11	93.45	95.33	96.30	95.71
	IMvGCN	**94.54**	95.31	95.81	95.84	95.42
	MMatch	93.41	95.10	95.73	95.81	96.05
	SPamCo	87.50	87.50	87.50	87.50	87.71
	TMLC	92.73	93.85	94.62	95.05	94.31
	Ours	94.53	**95.39**	**96.25**	**96.69**	**96.34**
	Δ	**-0.01**	**+0.03**	**+0.44**	**+0.39**	**+0.29**



(a) Error rate (b) Utilization rate

Fig. 3. Pseudo-label error rate and utilization rate during training on CIFAR-10 at $LR = 0.09$ and $IR = 0.1$. CDPL maintains a low error rate while retaining a high utilization rate.

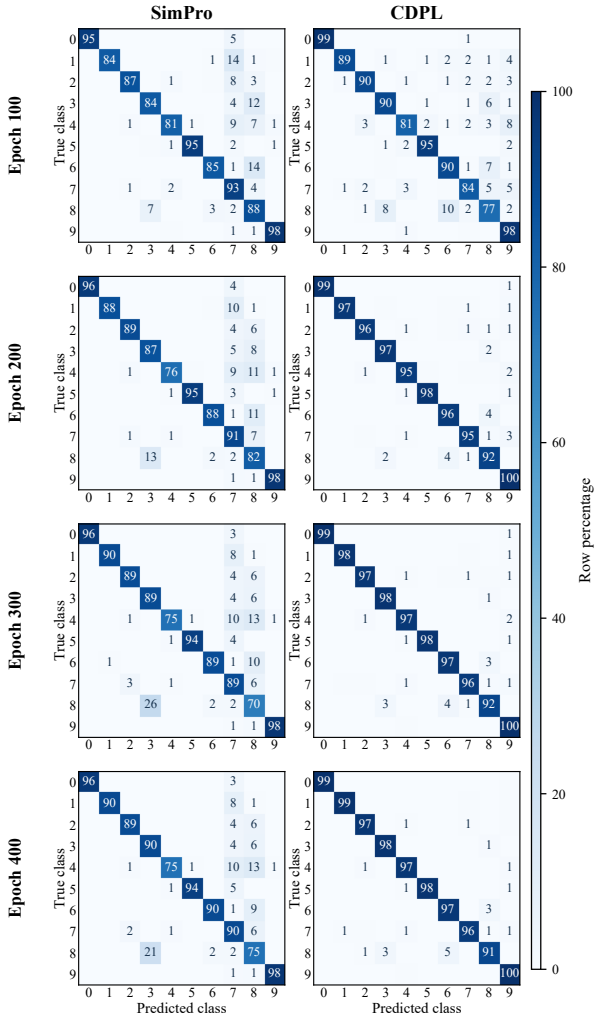
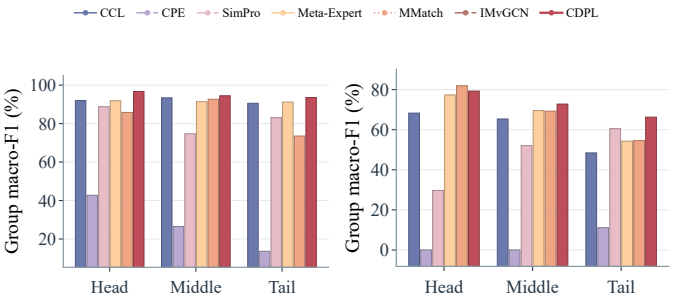
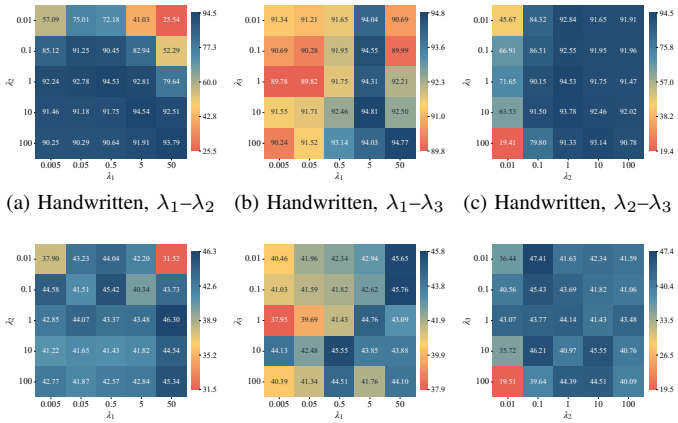


Fig. 4. Row-normalized pseudo-label confusion matrices of SimPro and CDPL on NoisyMNIST at $LR = 0.09$ and $IR = 0.1$.



(a) Handwritten (b) OutScene

Fig. 5. Grouped class-wise Macro-F1 across head, medium, and tail classes on Handwritten and OutScene under the consistent long-tailed setting with $LR = 0.09$ and $IR = 0.1$.



(a) Handwritten, λ_1–λ_2 (b) Handwritten, λ_1–λ_3 (c) Handwritten, λ_2–λ_3

(d) GRAZ02, λ_1–λ_2 (e) GRAZ02, λ_1–λ_3 (f) GRAZ02, λ_2–λ_3

Fig. 6. Parameter analysis of the trade-off coefficients λ_1, λ_2, and λ_3 on Handwritten and GRAZ02.

compact, well-separated class clusters, consistent with its strong and stable classification performance.

VIII. Additional Clinical Results on the IGF-1 Dataset

The main paper reports four metrics (Accuracy, Macro-F1, G-mean, and Specificity) for the pediatric IGF-1 clinical obesity-status classification task. Figure 9 reports the remaining five metrics, Precision, Recall, macro-averaged AUC (one-vs-rest), macro-averaged AUC (one-vs-one), and macro-averaged Average Precision (AP), under the same labeling ratios $LR \in \{0.06, 0.09, 0.12\}$. CDPL attains the best mean value on all five additional metrics, confirming that its advantage on the clinical task is not limited to the four metrics shown in the main paper.



(a) Consistent–Train (b) Inverse–Train (c) Arbitrary–Train (d) Uniform–Train



(e) Consistent–Test (f) Inverse–Test (g) Arbitrary–Test (h) Uniform–Test

Fig. 7. t-SNE visualization of the fused representations of CDPL on NoisyMNIST under four unlabeled data distributions (consistent, inverse, arbitrary, and uniform), with $LR = 0.09$ and $IR = 0.1$.

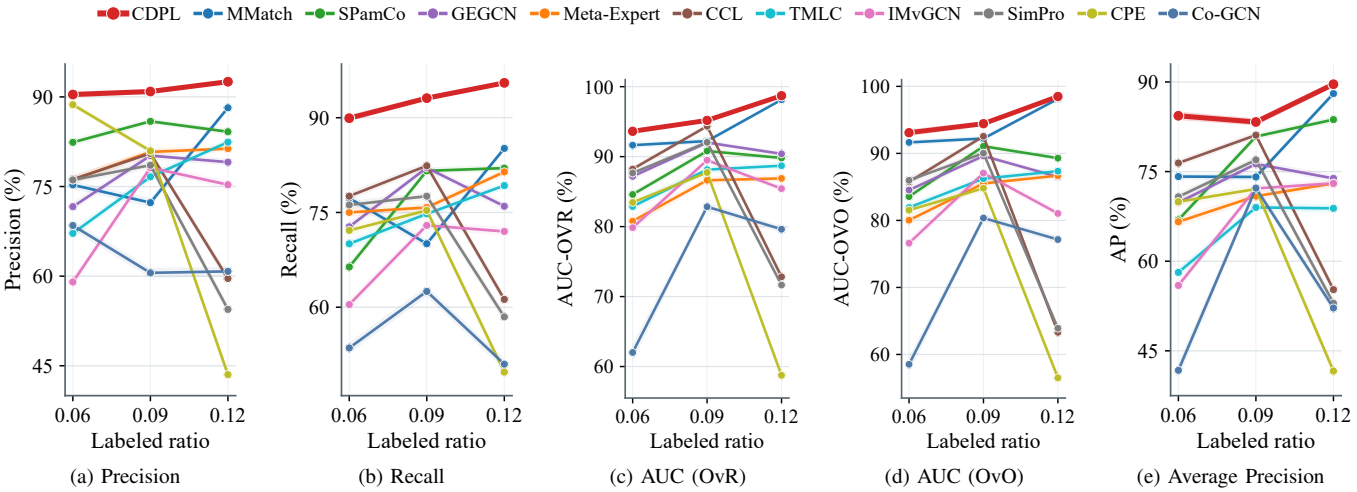


(a) Consistent–Train (b) Inverse–Train (c) Arbitrary–Train (d) Uniform–Train



(e) Consistent–Test (f) Inverse–Test (g) Arbitrary–Test (h) Uniform–Test

Fig. 8. t-SNE visualization of the fused representations of CDPL on YouTubeFace20 under four unlabeled data distributions (consistent, inverse, arbitrary, and uniform), with $LR = 0.09$ and $IR = 0.1$.



(a) Precision (b) Recall (c) AUC (OvR) (d) AUC (OvO) (e) Average Precision

Fig. 9. Precision, Recall, macro-averaged AUC (one-vs-rest and one-vs-one), and macro-averaged Average Precision on the pediatric IGF-1 dataset at $LR \in \{0.06, 0.09, 0.12\}$.